Sophie Zhao

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA — Expected May 2030
Bachelor of Science in Computational Physics, Intended Additional Major in Computer Science
Relevant Courses: Matter and Interactions (computational mechanics), Concepts of Mathematics & Proofs (discrete math)

Stuyvesant High School, New York, NY — June 2026

ACTIVITIES

Web App Developer - Scotty Labs Labrador — September 2026 - Present
- Conducted stakeholder interviews with users and campus staff about issues in the current room reservation system
- Collaborated with team members to ideate and design an improved reservation system using Figma and APIs
- Designed a room-allocation platform that uses AI-assisted recommendations to reduce campus congestion

Data Engineering Consultant - Design For America — September 2026 - Present
- Designing data organization and analytics solutions for an organization to analyze exam data of 4,000+ students
- Ideating a dashboard with a filtering system to query specific performance metrics and KPIs for grant reporting
- Conducted interviews and affinity diagramming to evaluate organization needs and determine design workflows

LEADERSHIP

Co-President - High School Astronomy Club — September 2023 - June 2026
- Raised $1500 to fund club telescope, binoculars, and trips to observatories and museums
- Collaborated with NASA ASTROCAMP and expanded club membership from 20 to 110+ members
- Awarded silver honor: 2025 International Astronomy & Astrophysics Competition (top 8% of 12,300 people)

Administrator & Tutor - Metis Project — September 2024 - June 2026
- Supervised 50 tutors and 200 students for NYC specialized high school entrance exam (SHSAT)
- Analyzed yearly aggregate diagnostic exam data to improve student performance
- Built a framework categorizing questions to isolate specific error distributions and deliver targeted feedback

PROJECTS

Smart Kitchen - NOVA Generative AI Hackathon — November 2026 - November 2026
- Designed a web platform for DoorDash using React and Vite to address delivery mistakes and scattered records
- Utilized SQLite transaction blocks in FastAPI to dynamically update menu availability based on inventory data
- Integrated OpenRouter LLM APIs to parse unstructured supplier receipts and generate owner profitability insights

VPython Gravity Assist Simulation — June 2026 - June 2026
- Recreated the gravity-assist sequences used by Voyagers 1 and 2, as well as a general mode that allows users to experiment with their own planetary and spacecraft configurations
- Implemented Runge-Kutta numerical integration and adaptive time-stepping in N-body gravitational simulator

SKILLS

Software: Jupyter Notebook, Web VPython, VS Code, LaTeX, Google Suite, FastAPI, Figma, GitHub
Computer Languages: Python, HTML/CSS, Java
Languages: English, Spanish, Chinese